Test-Time Score Distillation: Mask-Free Semantic Style Editing with Reverse-Process Regularization

Anonymous CVPR submission

Abstract

A style reference carries not only artistic appearance but also objects and composition, making it difficult to transfer its medium to a new subject. We cast training-free style transfer as test-time score distillation (TSD) on a frozen rectified-flow text-to-image transformer. At inference, we optimize only the current latent using the model's image-conditioned CFG-augmentation contrast, computed from velocity predictions with and without a style reference at the identical latent state and noise level. The corresponding score direction is RMS-normalized to construct a detached clean-state target, and latent updates are interleaved with frozen reverse sampling, which provides reverse-process regularization. TSD trains no weights and requires no attention-feature modification, additional adapter, fake-score network, or generator. Adjusting the reverse sampling and latent intervention schedules supports both semantic style editing with greater composition preservation and global style transfer with freer composition. Experiments across artistic media and target prompts show stronger reference style matching than the backbone's native image-and-text editing and direct reference guidance, while retaining recognizable target subjects.

1. Introduction

Style transfer on a frozen generator is treated as one task, but users want two. Given a content prompt and a reference image, a frozen text-to-image transformer can render the prompt's subject "in the style of" the reference. In practice two very different outcomes are desirable and are conflated under one name. In one, an artist wants the reference's *scene* preserved, including its composition and arrangement, with only the subject re-imagined in the reference's medium. This is *Semantic Style Editing*. In the other, the artist wants only the reference's *look*, including its medium, texture, and palette, carried onto a freshly composed scene, with the reference's specific layout thrown away. This is *Global Style Transfer*. The desired amount of reference-layout retention therefore depends on the creative task.

These two outcomes motivate a common *style-locality* axis. Style locality describes how much reference layout is retained. Greater retention supports semantic editing, while freer composition supports global transfer. We assess this axis separately from reference-medium fidelity and target-subject alignment, so replacing the reference subject remains an explicit requirement in either task.

We drive this axis with one frozen-model operator. Distribution-matching distillation separates CFG augmentation, which strengthens conditioning, from distribution matching with a learned fake-score estimate [19, 38]. Our setting differs: we optimize a single inference-time latent while retaining the frozen reverse process. The native reference-conditioning contrast supplies the style direction, and ordinary reverse steps continue the model's denoising trajectory between latent interventions. We call this procedure **Test-Time Score Distillation (TSD)** and its alternation *reverse-process regularization*.

TSD evaluates reference-conditioned and reference-free velocities at the same latent state, noise level, and target text. We convert their difference to a score direction, RMS-normalize it, and construct a detached clean-state target for latent optimization. The target is rebuilt after each inner update. All network weights remain frozen, with no learned style adapter, auxiliary fake-score network, or spatial mask. Sampling resolution and intervention settings jointly determine the style-locality operating point.

Contributions.

1. **A common task formulation.** Semantic Style Editing and Global Style Transfer are two goals on a style-locality axis. We serve both with one operator and study their sampling and latent-intervention settings.
2. **Test-time score distillation with reverse-process regularization.** We turn the frozen model's same-state reference contrast into an RMS-normalized pseudo-target for latent optimization, interleaved with its native reverse trajectory.
3. **A separable-axis evaluation protocol.** We evaluate style fidelity with CSD, subject fidelity with CLIP-T, and composition transfer with reference-layout similarity, interpreting all three together.

4. **Native-editor and dedicated-method comparisons.** We retain the initial 40-reference study and extend it with multiple prompts, a strong same-backbone reference-CFG control, and dedicated transfer methods. The joint readouts expose how guidance and intervention settings affect style, subject, and composition.

2. Related Work

Distillation and test-time score guidance. Distribution-matching distillation matches a student generator to a teacher through teacher and fake-score estimates, with later variants improving the objective and separating distribution matching from CFG augmentation [19, 38, 39]. This decomposition connects to diffusion and score models' noise-conditional vector fields [14, 30]. Conditional contrasts appear in classifier guidance, classifier-free guidance, score distillation, and text-guided editing [3, 6, 10, 13, 21, 23]. Other test-time methods steer frozen samplers with external losses, perturbed attention, or annealed conditions [1, 2, 15, 27]. TSD applies the reference-conditioned augmentation direction to one test-time sample. It optimizes the current latent and interleaves these updates with the frozen reverse trajectory.

Reference-conditioned style transfer and frozen-model editing. Classical style transfer matches feature statistics [8, 16]. Diffusion methods extend this idea through exemplar conditioning, learned style interfaces, attention sharing, or sampling-time transport [4, 5, 9, 12, 26, 28, 32, 35–37]. Related editing systems control cross-attention or reuse frozen diffusion features [11, 31], while rectified-flow methods develop inversion and direct trajectory editing [17, 25]. TSD belongs to the optimization-based, training-free-of-weights family, but its signal is the native same-state reference-conditioning contrast rather than replaced attention features, a trained adapter, or an explicit mask. We evaluate medium, palette, and texture transfer together with reference-layout retention, without requiring exact correspondence of individual motifs.

Dedicated transfer systems and latent optimization. Attention Distillation is a close optimization-based comparison: it updates the sampling latent using an attention-derived reference loss [40]. StyleShot uses a style-aware image encoder [7], CSGO learns content–style composition [34], and USO supports style and subject customization [33]. TSD uses the native reference-conditioned velocity contrast as its supervision. Our comparisons distinguish reference-plus-text generation from methods that additionally require a content image.

3. Method

Figure 1 summarizes the operator derived in this section as an alternating cycle of task-direction distillation, latent optimization, and frozen-prior regularization. The reference contrast supplies the task direction, an RMS-normalized pseudo-target defines the latent objective, and the reverse process advances the sample between interventions.

3.1. Task and native reference conditioning

Let r denote the reference image and c the full target instruction, including the requested subject and transfer instruction. We consider two objectives. *Semantic style editing* seeks the requested subject in the reference medium with substantial reference-layout retention. *Global style transfer* seeks the medium in an independently composed image. In both cases, the target text specifies the output subject, while the reference supplies the medium and, to a task-dependent degree, layout.

The frozen transformer predicts a velocity at latent z_σ and noise level σ. We query its native conditioning interface twice:

$$\begin{aligned} v_r &= v_\theta(z_\sigma, \sigma, c, r), \\ v_b &= v_\theta(z_\sigma, \sigma, c, \varnothing). \end{aligned} \tag{1}$$

In the Klein implementation, the frozen VAE encodes the reference into auxiliary image tokens. The transformer receives these tokens together with the current noisy-image tokens and target-text representation. Image-position identifiers distinguish the reference tokens from the output tokens. The reference-free branch omits the auxiliary image tokens. Text, latent state, noise level, and output geometry are shared between the two queries, so their difference measures the response to adding the reference condition.

3.2. From velocity contrast to a score target

We express the native conditioning contrast in score space using the rectified-flow interpolation $z_\sigma = (1-\sigma)x_0 + \sigma\epsilon$, with Gaussian ϵ and velocity convention $v = \epsilon - x_0$. The clean estimate and score satisfy

$$\hat{x}_0 = z_\sigma - \sigma v, \tag{2}$$

$$s_\theta = -\frac{z_\sigma + (1-\sigma)v_\theta}{\sigma}. \tag{3}$$

Equation 3 follows from the Gaussian-noise score identity with the conditional-mean velocity. A learned field approximates that quantity [18, 20]. The shared state cancels when the two conditions in Eq. 1 are contrasted:

$$\Delta s_\sigma = s_r - s_b = -\frac{1-\sigma}{\sigma}(v_r - v_b). \tag{4}$$

The factor $-(1-\sigma)/\sigma$ fixes the orientation of the score direction under this velocity convention. We clamp the denominator near zero and apply optimization only at positive noise levels.

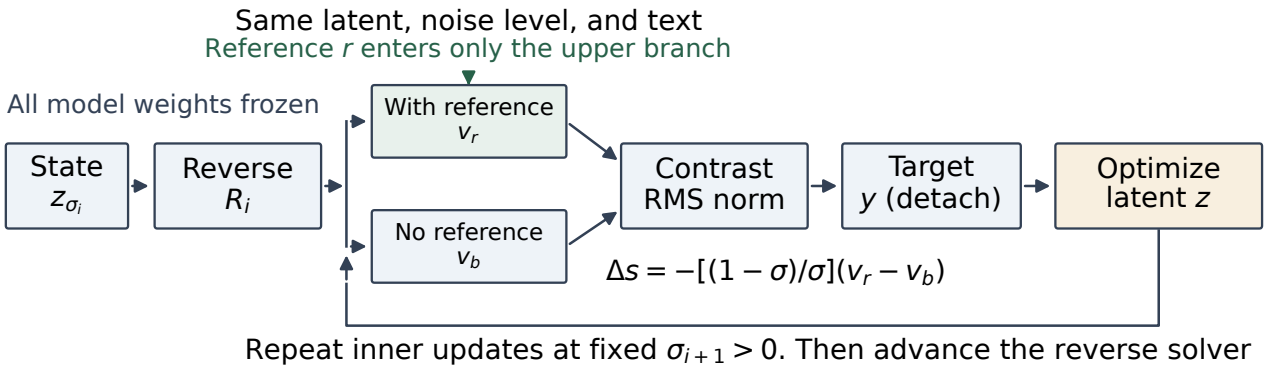

Figure 1. TSD inference. The frozen reverse step reaches σ_{i+1}, where paired reference-conditioned and reference-free predictions define an RMS-normalized score direction. A detached target then guides inner latent optimization at the reached noise level. The cycle repeats with all network weights frozen.

RMS normalization makes the target offset insensitive to most changes in field magnitude across noise levels. For a positive multiplier a and normalization floor ε_g, define

$$g = a\Delta s_\sigma, \qquad d = \frac{g}{\max\left(\sqrt{N^{-1}\sum_{j=1}^{N} g_j^2}, \varepsilon_g\right)}, \tag{5}$$

where N counts latent elements per sample. Away from the numerical floor, the positive multiplier a cancels under normalization. The resulting direction has unit RMS, giving the target offset a consistent scale across noise levels.

3.3. RMS-normalized pseudo-target distillation

The pseudo-target supplies a first-order optimization signal without backpropagating through the frozen transformer. With sg denoting stop-gradient, each inner iteration forms

$$\hat{x}_0(z) = z - \sigma\,\mathrm{sg}(v_b), \tag{6}$$

$$y = \mathrm{sg}(\hat{x}_0(z) + \eta d)\,, \tag{7}$$

$$L_{\mathrm{p}} = \frac{1}{2N}\sum_{j=1}^{N}(\hat{x}_{0,j}(z) - y_j)^2, \tag{8}$$

where η sets the normalized target offset. Both teacher predictions used to construct d are detached and recomputed at the current inner-loop state. The target is rebuilt before each loss evaluation, then treated as a constant during differentiation.

Detaching the velocity makes the clean-estimate Jacobian the identity in this branch. Thus

$$\frac{\partial \hat{x}_0}{\partial z} = I, \qquad \nabla_z L_{\mathrm{p}} = -\frac{\eta}{N} d. \tag{9}$$

The mean reduction contributes $1/N$. For SGD with learning rate ℓ, the displacement is $\ell\eta d/N$. We use AdamW, whose moment estimates and numerical stabilization determine the displacement from this gradient. Thus η controls the target offset and ℓ controls the optimizer step. The supplementary derivation gives both update expressions.

The evaluated TSD operator uses L_{p} as its latent objective. We initialize AdamW with zero weight decay at each outer intervention and retain its moments across the inner iterations at that node. Recomputing the target after every update keeps the reference contrast evaluated at the current state. No warm-start or output AdaIN transformation is used.

Why a directional target? The target uses the direction of the reference contrast rather than a loss that forces the two predictions to agree. Minimizing $\frac{1}{2}\|\Delta s(z)\|^2$ would differentiate the contrast itself and produce $J_{\Delta s}(z)^\top \Delta s(z)$. Our detached target instead gives the first-order direction in Eq. 9, without differentiating the teacher. Normalization keeps the target offset informative even when the contrast magnitude changes along the trajectory. Rebuilding the target after each update lets successive interventions follow the reference response at the edited state.

3.4. Reverse-process regularization and style locality

The reference contrast provides a task direction, while the reverse field provides the generative dynamics. This separation is motivated by the CFG-augmentation and distribution-matching components of decoupled DMD [19]. Their roles occur in different optimization problems: DMD updates generator weights using a learned estimate of the student distribution, whereas TSD updates one latent inside an existing multi-step sampler. We retain that sampler as

Algorithm 1 TSD inference

Require: Frozen v_θ, instruction c, reference r, solver nodes, interventions $\mathcal{G}$, inner counts K_i, η, optimizer rate ℓ

Initialize Gaussian latent z_{σ_0}
for $i = 0, \dots, M-1$ **do**
  $z \leftarrow R_i(z_{\sigma_i})$ at σ_{i+1}
  if $\sigma_{i+1} \in \mathcal{G}$ and $\sigma_{i+1} > 0$ **then**
    Initialize latent optimizer at z
    for $k = 1, \dots, K_i$ **do**
      Evaluate detached v_r, v_b at the same (z, σ_{i+1}, c)
      Construct d, y, and L_{p} using Eqs. 5–8
      Update only z using $\nabla_z L_{\mathrm{p}}$
    end for
  end if
  $z_{\sigma_{i+1}} \leftarrow z$
end for
Decode z_{σ_M}

the regularizing component of the inference procedure. No learned student-distribution estimate is needed to define or execute the TSD operator.

We implement reverse-process regularization by alternating reference-directed latent optimization with the frozen model's text-conditioned reverse steps. The reverse solver advances the state before a guidance intervention. Let $\sigma_0 > \cdots > \sigma_M = 0$ be its actual noise nodes and R_i its reverse update. From z_{σ_i}, we first obtain $z^{(0)}_{\sigma_{i+1}} = R_i(z_{\sigma_i})$. If the reached positive node belongs to the intervention set $\mathcal{G}$, we apply K_i latent optimization steps there. At other nodes we retain the solver output. Algorithm 1 specifies this order. The final reverse step has no subsequent inner optimization.

Style locality and working points. The solver nodes determine where reverse sampling evaluates the model, while $\mathcal{G}$, K_i, η, and ℓ determine the latent interventions. We use their joint settings to obtain different degrees of style strength and reference-layout retention. The retained configurations use 25 reverse intervals with two inner updates and $\eta = .05$, or eight intervals with five inner updates and $\eta = .12$. Both use learning rate .01. Inner updates occur after eligible positive reverse nodes. For controlled comparisons of discretization, we also insert shared positive nodes and assign the same inner count at each node. This separates solver resolution from intervention location and count.

3.5. Relationship to CFG and computational cost

Text guidance and reference guidance act on different conditions. Let v^β_u denote text CFG applied with image condition u, using $v_{\mathrm{uncond}} + \beta(v_{\mathrm{cond}} - v_{\mathrm{uncond}})$ within that branch. Direct reference guidance advances the reverse solver with

$$v^\beta_{\mathrm{refCFG}} = v^\beta_b + \gamma(v^\beta_r - v^\beta_b). \tag{10}$$

Both image conditions use the same text scale β. In the retained TSD settings, the reverse step uses the reference-free, text-guided velocity v^β_b. Reference information enters through the inner score-contrast target at fixed instruction c. The sampler text scale β, direct reference scale γ, and score multiplier a are recorded separately.

Computation. Each inner iteration evaluates the two frozen teacher branches and backpropagates through the detached-target loss to the latent. The transformer is evaluated under stop-gradient, so its activations need not be retained for latent differentiation. With $U = \sum_i K_i$ inner iterations, an uncached implementation adds $2U$ teacher evaluations to reverse sampling. Prediction sharing can reduce this count. The cost analysis records model calls and latent backward calls. The final latency and memory comparison will use common hardware with explicit text-guidance and offload settings.

4. Experiments

We evaluate TSD along the three axes introduced above: reference medium, requested subject, and reference-layout retention. The initial study tests both task regimes against the native editor on 40 references. An expanded study adds multiple prompts, dedicated style-transfer methods, and stronger same-backbone guidance controls. We then examine how latent interventions change the balance between style and subject fidelity.

4.1. Experimental setup

Initial and expanded studies. The initial study uses 40 style references with the prompt "a panda eating bamboo" and seed 65. It compares the original semantic and global TSD configurations with native editing instructions that respectively request reference-layout retention and a new composition. We retain its stored measurements as a distinct study. The expanded study uses newly generated outputs with recorded configurations and input/output hashes.

The new reference collection contains 48 public Krea-generated images across eight medium categories. We selected references for medium clarity, visual quality, and diversity before inspecting generated outputs. Each category contributes one development and five test references, giving an 8/40 split. Every method receives the same RGB 512×512 reference, obtained by aspect-preserving resizing and center cropping of the public 1024-pixel source version. Six fixed prompts request a panda, castle, cat, sunflower field, fisherman portrait, and lighthouse. Source identifiers,

transforms, and image hashes are provided in the supplementary records. The completed expanded comparison uses the eight development references and seed 64. The 40 disjoint references are reserved for the multi-seed test protocol in Sec. 4.5.

Comparison methods. The common task supplies a style reference and target text. TSD and reference CFG share the frozen Klein-9B-base backbone, text CFG 4, and the same transfer instruction. The reference-CFG baseline is chosen from a development sweep of guidance strengths and sampling budgets. Its retained setting uses 72 intervals and reference scale $\gamma = 5$. Attention Distillation (AD) uses SD1.5 with 50 steps, two inner updates, learning rate .015, text CFG 7.5, and AdaIN [40]. USO uses its style-only interface with FLUX.1-dev, pretrained rank-128 adapters and SigLIP features, 25 steps, and guidance 4 [33]. USO stores weights in FP8 with bf16 computation. Backbones, pretrained adapters, and precision are recorded alongside the method settings. StyleShot uses its pretrained SD1.5 style encoder and adapter with style scale 1, text CFG 7.5, and the official float32 pipeline [7]. Its nominal 50-step PNDM schedule performs 51 UNet evaluations. InstantStyle uses the official style-block IP-Adapter interface on SDXL, with 30 Euler steps, text CFG 5, style scale 1, and float16 inference [32]. Its image condition enters the designated style block. Quality and negative prompts follow the official example and are recorded in the supplement.

Metrics and paired analysis. We report CSD reference–output similarity for style [29], CLIP ViT-B/32 image–text similarity to the literal content request [24], and DINOv2 ViT-B/14 reference-layout similarity [22]. For normalized patch features P_x, let $q_x = \mathrm{vec}_{\mathrm{off}}(P_x P_x^\top)$. The layout measure is

$$\Lambda(x, r) = \mathrm{corr}(q_x, q_r). \tag{11}$$

Higher Λ indicates more similar spatial self-similarity structure, which can be useful for semantic editing. Both geometry and shared semantics can affect this readout. We compute method differences on matched reference/prompt/seed cells and bootstrap reference groups with all their repeated observations together, using 10,000 resamples.

4.2. Full benchmark against the frozen model's native edit

The same frozen model's native image-and-text editing path provides the initial baseline. In TSD, the reference enters the score-contrast objective while the reverse solver itself uses the reference-free branch. Setting the contrast to zero therefore gives text-only sampling, rather than native image editing. The native keep-layout and new-layout instructions are separate reference-conditioned comparisons.

Table 1. Initial 40-reference study, one content prompt and seed 65. Values are mean ± sample SD across references. S/G are the original semantic/global configurations. Layout measures retention of reference structure, whose desired level depends on the task.

Method	CSD ↑	CLIP-T ↑	Layout
TSD-S (semantic)	0.719 ± 0.079	0.295 ± 0.031	0.572 ± 0.148
TSD-G (global)	0.615 ± 0.096	0.304 ± 0.027	0.414 ± 0.109
Native keep-layout	0.267 ± 0.165	0.319 ± 0.012	0.317 ± 0.073
Native new-layout	0.247 ± 0.157	0.315 ± 0.011	0.299 ± 0.065

The initial study tests whether one operator can strengthen the reference medium under both composition objectives. Table 1 retains the original 160 measurements, with sample SD across the 40 references. The table summarizes the stored measurements from the initial study. Its complete per-reference records appear in the supplement.

TSD-S increases mean CSD from .267 to .719 relative to native keep-layout editing, and TSD-G increases it from .247 to .615 relative to native new-layout editing. Every reference has a positive paired CSD difference in both comparisons. CLIP-T decreases from .319 to .295 and from .315 to .304, respectively. The two TSD configurations also differ in layout retention: .572 for S and .414 for G. Thus the original study supports two useful operating regimes, with stronger reference matching accompanied by different amounts of reference structure.

CLIP-T is read against the literal content request, separately from the style-transfer instruction. The earlier five-reference, four-additional-prompt study is retained in the supplement using its stored prompt-level summaries. The new multi-prompt comparison below has its own inputs and measurements.

The style increase must be interpreted together with composition. Against each output's paired native baseline, only 1/40 semantic and 3/40 global outputs have no greater layout similarity. The full-set CSD gains therefore do not by themselves establish a style advantage at matched layout. The expanded study below evaluates this condition explicitly and includes a swept reference-guidance control.

4.3. Expanded reference-plus-text comparison

We select TSD settings and direct-reference controls on a 16-cell grid of four references, two prompts, and two seeds. The resulting 25-configuration study contains 400 outputs. We then compare the retained settings, AD, USO, StyleShot, and InstantStyle on all eight development references and six prompts at seed 64, giving 48 paired inputs per method. Table 2 reports all 336 outputs in this comparison. The supplementary material gives the full configuration study, including exact reused cells.

Both TSD settings strengthen style matching over the

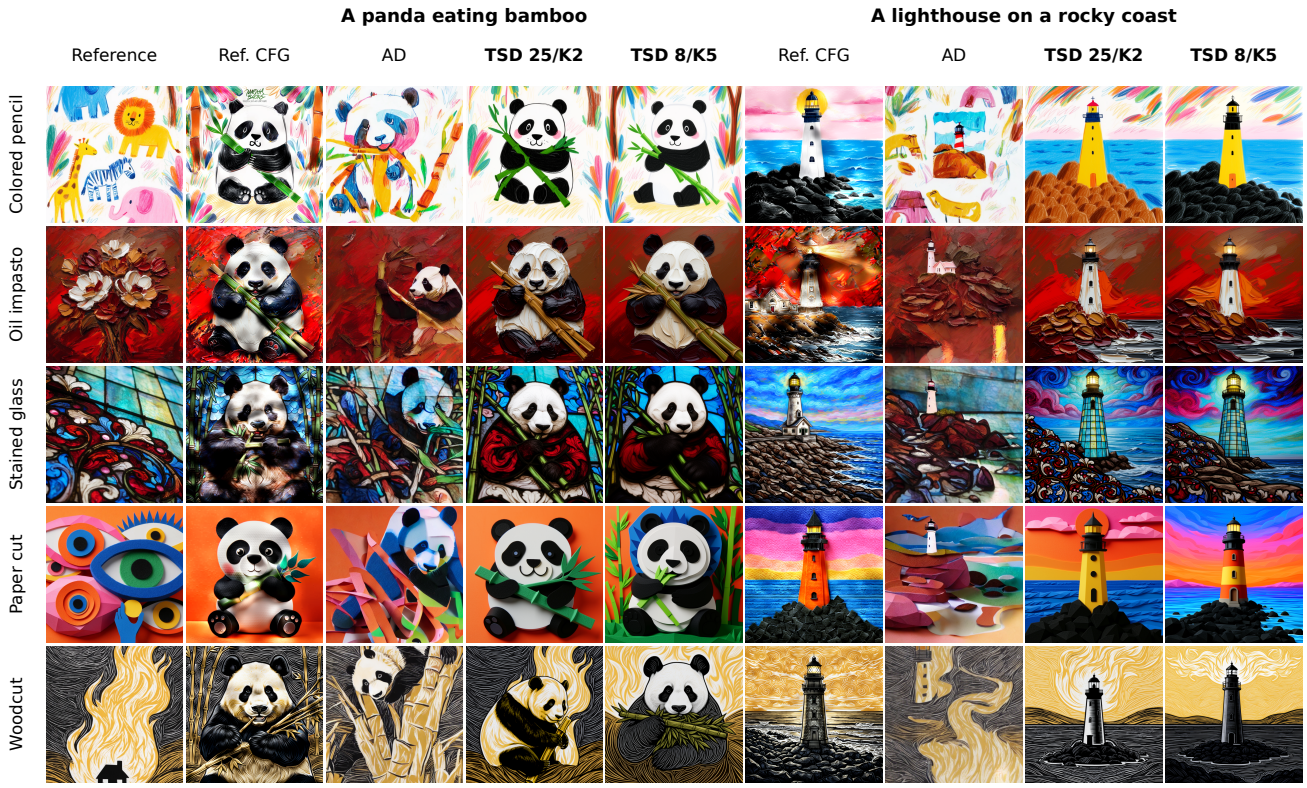

Figure 2. Reference-guided generation across two content prompts and five media. Each row shares a reference. Each prompt block compares reference CFG, Attention Distillation, and two TSD settings on the same input and seed 64. TSD preserves characteristic color, material, and surface patterns across the two subjects. The 25/K2 and 8/K5 settings change the strength and spatial expression of the reference signal. These outputs belong to the expanded development comparison in Table 2.

Table 2. Development coverage on eight references, six prompts, and seed 64 (48 outputs per row). Layout measures reference-layout retention. Backbones and precision are specified in the setup.

Method	CSD ↑	CLIP-T ↑	Layout
Reference CFG, $72/\gamma = 5$	0.3190	0.3005	0.3118
AD (SD1.5)	0.7069	0.2692	0.3224
USO (FLUX.1, FP8)	0.4117	0.3178	0.2953
StyleShot (SD1.5)	0.5042	0.2840	0.2703
InstantStyle (SDXL)	0.4947	0.2616	0.2900
TSD, 8/K5	0.4291	0.3078	0.3101
TSD, 25/K2	0.5751	0.2982	0.3503

selected reference-CFG baseline. TSD8/K5 raises mean CSD from .3190 to .4291 and CLIP-T from .3005 to .3078. The paired differences are .1101 for CSD (reference-group 95% bootstrap interval $[.0720, .1463]$) and .0073 for CLIP-T ($[.0004, .0157]$). Its mean layout similarity is .3101, compared with .3118 for the direct control. TSD25/K2 further increases CSD to .5751, with CLIP-T .2982 and layout similarity .3503. Its paired CSD increase over reference CFG is .2561 ($[.2178, .2940]$).

The dedicated methods place these settings in a broader style–subject comparison. Relative to AD, TSD25/K2 has higher CLIP-T by .0291 ($[.0233, .0354]$) and lower CSD by .1319 ($[.0559, .2049]$ for AD minus TSD). Relative to USO, it has higher CSD by .1633 ($[.0876, .2506]$) and lower CLIP-T by .0195 ($[.0149, .0249]$ for USO minus TSD). TSD8/K5 favors subject alignment within our two settings, while TSD25/K2 strengthens the style signal. StyleShot obtains CSD .5042 and CLIP-T .2840, compared with .5751 and .2982 for TSD25/K2. InstantStyle obtains CSD .4947 and CLIP-T .2616. These development comparisons use eight reference groups.

Figure 2 shows how the reference signal appears in different media and subjects. In stained glass, TSD carries colored panes and dark outlines into both the panda and lighthouse. The paper-cut examples retain layered shapes, while the oil-paint examples reproduce broad impasto strokes. The two TSD settings also differ visibly in framing and surface treatment. These examples complement the joint metric readout: medium transfer and reference-layout retention

are related properties, but neither is a substitute for rendering the target subject.

Paired layout constraint. We also evaluate style differences on cells whose TSD output retains no more layout than its corresponding baseline output:

$$\mathcal{I} = \{i : \Lambda(x_i^{\mathrm{TSD}}, r_i) \leq \Lambda(x_i^{\mathrm{baseline}}, r_i)\}. \quad (12)$$

Against reference CFG, TSD8/K5 satisfies this condition on 22/48 cells. Within this subset, its mean CSD and CLIP-T differences are .0867 and .0041. TSD25/K2 satisfies the condition on 18/48 cells, with differences .2127 and $-.0045$. We report these descriptive subset results together with their coverage and the complete-grid comparison.

4.4. The style-locality axis and ablations

Why use a moderate update count? At a fixed 25-step solver, learning rate .01, and target offset .05, K1/K2/K5 produce CSD .3726/.5555/.7302 and CLIP-T .3109/.3110/.2796 on the same 16-cell search grid. K2 strengthens style relative to K1 while retaining a similar subject score, and uses 146 DiT forwards compared with 290 for K5. This motivates the retained 25/K2 setting. Reducing the K5 learning rate to .005 instead gives CSD .6288 and CLIP-T .2986. Update count and learning rate therefore provide different style–subject operating points.

Sampling and intervention location. The common-node controls insert $\{.8, .6, .4, .2\}$ and perform one update at each positive node. This fixes guidance locations and counts across nominal solver lengths. The dense configurations intervene at the original positive solver nodes. We record actual intervals, optimizer choice, update count, and displacement for each configuration. The complete search also includes RMS-SGD and shorter reference-guidance controls. At the four shared nodes, nominal 25-step and eight-step sampling yield CSD .0386/.0315 and CLIP-T .3190/.3168, respectively. Both have much lower style similarity than the dense interventions in this study. These measurements show the importance of the intervention schedule alongside solver resolution. This design compares discretization with matched intervention locations. The dense 8/K5 and 25/K2 settings instead change several factors together and define operating points rather than an isolated effect of reverse-step count.

Model calls and timing. TSD8/K5 and TSD25/K2 use 86/146 DiT forwards and 35/48 latent backward calls respectively. The backward calls differentiate the detached-target loss. Reference CFG uses 288 DiT forwards, AD uses 200 UNet forwards, and USO uses 25 DiT forwards under their respective settings. StyleShot uses 51 UNet forwards.

Table 3. Held-out comparison, results pending. Each row has a planned 720 cells on the same 40-reference, six-prompt, three-seed grid. TBD denotes an unfilled result, including rows whose generation may already be complete.

Method	CSD ↑	CLIP-T ↑	Layout
Reference CFG	*TBD*	*TBD*	*TBD*
AD	*TBD*	*TBD*	*TBD*
USO	*TBD*	*TBD*	*TBD*
StyleShot	*TBD*	*TBD*	*TBD*
InstantStyle	*TBD*	*TBD*	*TBD*
CSGO	*TBD*	*TBD*	*TBD*
TSD 8/K5	*TBD*	*TBD*	*TBD*
TSD 25/K2	*TBD*	*TBD*	*TBD*

Calls from different backbones describe different computations. A common-hardware timing comparison is reserved below. End-to-end timing includes reference/text encoding, sampling, latent optimization, and output decoding. Model loading is measured separately, and single- and two-GPU placements are grouped separately in cost records.

Inference cost (pending). Complete the same-hardware comparison of end-to-end latency, peak allocated memory, precision, and offload policy. Separate single- and two-GPU placements and report model loading separately. The measured model-call counts above remain part of the completed analysis.

4.5. Multi-seed comparison

The held-out protocol uses 40 disjoint test references, six target prompts, and seeds 64/65/66, giving 720 reference/prompt/seed cells per configuration. The first frozen matrix contains reference CFG, AD, USO, and two TSD settings, or 3,600 outputs. A separately validated extension adds StyleShot, InstantStyle, and CSGO on the same inputs. The final comparison thus reserves six baselines and two TSD settings, with a planned total of 5,760 outputs. Table 3 provides the result slots. No test result is used to select the method settings.

Held-out results (pending). Complete generation, metric evaluation, and reference-group paired intervals for the original matrix and validated baseline extension. Fill Table 3, followed by the paired layout-constrained analysis and per-medium comparison.

4.6. Perceptual evaluation

The planned blinded study pairs TSD25/K2 with each of the six baselines on 40 fixed reference/prompt/seed cases. Each pair is assigned three independent human judgments. The target is 240 pairs and 720 judgments, with separate questions on style, subject, and overall preference. The

displayed method names are hidden and left/right positions are randomized. The supplementary protocol specifies sampling and uncertainty estimation.

Human preferences (pending). Collect the planned judgments and report actual participant and response counts, wins, ties, unavailable responses, and reference/rater-aware uncertainty. Overall preference is the primary readout. No human preference result is available in this draft.

5. Discussion

TSD makes the frozen model's response to a reference an explicit objective for the image latent. The reference signal and reverse sampling share one model, and the intervention schedule provides different balances of style strength, subject alignment, and layout retention. These properties make TSD applicable to both composition-preserving semantic editing and freer style-conditioned generation through a common operator.

The scope of the transfer follows the model's native reference representation. A reference can convey object identity and layout as well as material and texture, so the desired operating point depends on the creative task. The current collection contains synthetic references, and extending the evaluation to photographed materials and human-made artworks would broaden the assessed range. Repeated teacher evaluations also make inference cost an important design consideration. Prediction reuse and adaptive intervention schedules are promising directions for reducing it.

6. Conclusion

Training-free style transfer on a frozen transformer serves Semantic Style Editing and Global Style Transfer through a common style-locality framework. TSD applies the native reference-conditioning score direction through RMS-normalized detached targets for latent optimization. Interleaving those updates with the frozen reverse trajectory provides reverse-process regularization, without training a generator or auxiliary fake-score network. The initial 40-style study and expanded comparisons show stronger reference matching than native editing and direct reference guidance. Sampling and intervention settings provide different balances of medium, subject, and composition, with no mask or segmentation.

LLM Usage Statement

OpenAI Codex assisted manuscript drafting and editing, mathematical and code checks, protocol and experiment-script development, reference curation, qualitative inspection, experiment management, and figure/statistical-analysis code. The new reference images are Krea-generated. Numerical results are computed from recorded executions. The authors are responsible for verifying the text, references, data provenance, and computations.

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